DATA ANALYTICS PROJECT REPORT

Bloomberg P&L Data Extraction

HEIA NA Equity, FY2017–FY2024

Analysis and visualization of income statement data (EUR mn)

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Tools: Python (pandas, NumPy, openpyxl), Excel, Power BI

Structure: Stages 0 to 9, CFO memo, final reflection

Stage 0: Project Goal and Problem Statement

Q1. Goal, decision-maker and decision

The goal is to turn an eight-year Bloomberg income statement for HEIA NA Equity (a global brewer, FY2017 to FY2024, EUR mn) into a clean, reconciled, decision-ready P&L model, dashboard and memo. The decision-maker is the **CFO**; an external equity analyst is the secondary reader. The output helps the CFO decide where profitability is leaking (gross margin, distribution and logistics, items below EBIT) and whether the recovery after FY2020 is sustainable. That in turn drives pricing, cost and capital-allocation priorities, including the dividend.

Q2. Why presentation quality is an analytical risk

Text-formatted numbers silently drop out of sums, mixed units distort totals by orders of magnitude, and a wrong sign flips a subtotal. A manager who reads a distorted statement draws the wrong conclusion without any visible error.

Example: excise tax is printed as a positive number (EUR 3,910 mn in FY2017) while every other deduction is negative. Added to Total Revenue it gives EUR 29,700 mn instead of the correct Net Revenue of EUR 21,880 mn, and every margin is then built on the wrong base (FY2017 gross margin would read 38.2% instead of 51.8%).

Q3. Objectives

1. Audit the extract and log every defect before changing anything.
2. Clean and reshape the data in Python (numeric conversion, duplicates, unit and sign fixes) and prove it with reconciliation checks.
3. Build a lookup-driven master P&L and ratio sheets in Excel.
4. Analyse trend, margin, volume/price and segment performance.
5. Present the results in a Power BI dashboard and a one-page CFO memo.

Q4. Role of each tool

Python cleans, reconciles and exports one tidy dataset, so the work is repeatable and testable. **Excel** gives the transparent, lookup-driven model with formatting and charts that a finance reader can audit cell by cell. **Power BI** gives interactive filtering and storytelling for a non-technical audience. Python alone is hard to audit, Excel alone does not clean messy text reliably, and Power BI alone does not reconcile the data. Together they give a defensible view rather than an impression.

Q5. Three events visible in FY2017 to FY2024

- **The pandemic (FY2020):** inspect On-Trade revenue in the channel split first, then Net Revenue.
- **Input-cost and energy inflation (FY2022):** inspect Cost of Sales and gross margin first.
- **The Russia exit (FY2022 to FY2023):** inspect Net Income from Discontinued Operations, which sits below EBIT.

Q6. Excise duty and net revenue

Excise duty is a tax on alcohol that the brewer collects and passes to the government. **Total Revenue** includes it and **Net Revenue** excludes it. Margins are built on net revenue because excise is a pass-through that varies by country and product mix, so it says nothing about the company's own performance and a larger denominator would hide real margin changes.

Stage 1: Audit of the Extract

Q1. Sheets, fiscal years and line items

Sheet	What it holds
HEIA NA Equity	Raw standardised income statement, FY2017 to FY2024
REV BY SEGMENT	Net revenue by product category, brand and distribution channel
REV BY GEO	Net revenue by region under the old and the new segmentation
OPEX DETAIL	Expense lines in EUR and as % of net revenue, plus volume, employees and per-hl metrics
LOOKUP TABLES	Line-item dictionary, region mapping table and extract metadata
README	Notes on the extract

There are **8 fiscal years** and **34 distinct line items** on the income statement (excluding the duplicate Net Revenue row and the four ratio rows: growth, gross margin, operating margin, effective tax rate).

Q2. Where the data begins

Rows 1 to 4 hold the title, ticker, statement type and units. The two header rows are rows 7 and 8, and data starts at row 10. Footnotes sit below the table (rows 57 to 62). No tool can read the file as-is because of the title block, the two-row header, indented labels, footnote markers in column B, text numbers and placeholders.

Q3. Single-header import

The period labels would come from one row only. The dates row (12/31/2017 and so on) becomes the first data row, every column is typed as text, and the FY labels and dates cannot be combined into one clean column name.

Q4. Values that look like numbers but are not

- Thousands separators stored as text: “21,880.0”.
- A non-breaking space as separator: “(2\xa0550.0)” (Personnel Expenses, row 27).
- Padding spaces: “ 10,870.0 ” (Off-Trade, row 12).
- A Unicode minus: “−40.0” (Other Net Finance, FY2018, row 40).
- Parentheses for negatives: “(10,546.0)”.
- Percent signs: “2.7%”.
- A value in the wrong unit: “1,960,000” (Export & Other, FY2019).
- Placeholders: “—”, “#N/A N/A”, “NM”.

Q5. Negative-number styles

- **Parentheses:** cost of sales (row 21), operating expense lines and impairment (rows 27 to 33), interest expense and other finance (rows 39 and 40), tax (row 42), minority interest (row 47), net income attributable (row 48) and EPS such as (0.34) (rows 50 and 51).
- **Hyphen-minus:** “-75.0” (rows 44 and 46, FY2020), “-208.0” (row 46, FY2023) and the percent value “-17.7%” (row 18).
- **Unicode minus:** “−40.0” (row 40, FY2018).

Q6. The four placeholders

- **"—"** means nil or not reported (impairment in most years, discontinued operations in some). It is converted to 0 (a genuine nil, which is what makes EBIT tie) and flagged in a placeholder column.
- **"#N/A N/A"** means not available (discontinued operations in FY2017, FY2019 and FY2024; brand revenue in FY2023 and FY2024). It is kept as NaN, never zero.
- **"NM"** means not meaningful (FY2017 growth, which has no prior year). Kept as NaN.
- **"n.a."** means not available from the source (net revenue per hl in FY2017 on OPEX DETAIL). Kept as NaN, then derived as 21,880 / 218.0 = 100.4.

Counts: the income statement has 9 nil, 3 #N/A N/A and 1 NM placeholders; the segment sheet has 10 #N/A N/A (brand revenue, FY2023 and FY2024); OPEX DETAIL has 1 n.a. Deleting them all would be a mistake because "not available" and "zero" are different facts: turning brand revenue into zero would be false, and deleting rows would break the year alignment.

Q7. Subtotals and the duplicate row

Subtotals: Total Revenue (incl. Excise), Net Revenue, Gross Profit, Total Operating Expenses, Operating Profit (EBIT), Income Before Taxes, Net Income from Continuing Operations, Net Income (Total) and Net Income Attributable, plus the Total rows on the segment sheets. **Net Revenue appears twice** (rows 17 and 19) and Premium appears twice in the product block (REV BY SEGMENT, rows 7 and 11). Without tags, a sum over a column counts subtotals on top of their components and counts Net Revenue twice.

Q8. Unit error

Export & Other in FY2019 is 1,960,000 against 1,750 (FY2018) and 950 (FY2020). **Proof:** 9,740 + 11,890 + 380 + 1,960 = 23,970, which equals FY2019 Net Revenue, and REV BY SEGMENT shows 1,960.0. The raw value was in EUR thousand, so it is divided by 1,000 (footnote 2 asks for exactly this check).

Q9. Excise sign convention

Excise is shown positive (3,910 in FY2017) while every other cost is negative. Total Revenue minus Excise reconciles to Net Revenue in every year (FY2017: 25,790 − 3,910 = 21,880 mn). Adding it would give 29,700. It is tagged as a Negative-sign deduction in the clean data.

Q10. OPEX DETAIL total versus the raw statement

The totals differ in **FY2020** (9,220 vs 8,320, a gap of EUR 900 mn) and **FY2023** (12,400 vs 11,350, a gap of EUR 1,050 mn). The gaps equal the impairment of PP&E and goodwill: OPEX DETAIL includes impairment in its total, while the raw statement shows it as a separate line below Total Operating Expenses. Check: FY2020 EBIT = 9,840 + 64 − 8,320 − 900 = 684.

Q11. Data-quality log

Full log: data_quality_log.csv (20 entries). Row numbers are Excel rows of the source workbook.

Issue	Sheet and row	Evidence	Fix
Unit error	HEIA NA Equity, row 14 (FY2019)	1,960,000 against 1,750 and 950	Divide by 1,000
Excise sign	HEIA NA Equity, row 16	Positive while costs are negative; 25,790 − 3,910 = 21,880	Tag as deduction; verify TR − excise = NR

Issue	Sheet and row	Evidence	Fix
Duplicate rows	Rows 17 and 19; REV BY SEGMENT rows 7 and 11	Identical label and values in all years	Drop one, flag it
Impairment inside OPEX total	OPEX DETAIL, total row	9,220 and 12,400 against 8,320 and 11,350	Keep impairment as a separate line
Placeholders	Rows 18, 33, 45; OPEX DETAIL row 26	"—", "#N/A N/A", "NM", "n.a."	Convert to NaN or nil; derive the n.a. value
Unicode minus	Row 40 (FY2018)	"−40.0"	Replace with "-"
Non-breaking space	Row 27	"(2\xa0550.0)"	Replace \xa0
Text numbers, padding, % signs	Whole statement	Commas, spaces and % stored as text	to_number()
Label mismatches	Sheets vs dictionary	& vs and; Amortisation vs Amortization; trailing spaces	Normalise labels
Two codes missing	LOOKUP TABLES, rows 17 and 23	Distribution & Logistics and Share of Associates & JVs have no code	Graceful fallback in lookup; provisional codes in Python
Region restatement	REV BY GEO	Two segmentations; Central & Eastern Europe split	Mapping table with allocation %

Stage 2: Data Cleaning and Restructuring (Python)

Q1. Reading the file

header=None (or an explicit skiprows) stops pandas guessing the header from the title block and lets me control both header rows. I read the workbook with dtype=object and keep_default_na=False, so pandas cannot silently turn "#N/A N/A" into NaN or convert text to floats. For the CSV mirror I use encoding="utf-8-sig", because the data contains non-ASCII characters (the Unicode minus and the trademark sign in a brand name); it also strips a byte-order mark if present. The cleaned CSVs are written with utf-8-sig so Excel opens them correctly.

Q2. One conversion function

```
def to_number(x):
    if x is None or (isinstance(x, float) and np.isnan(x)): return np.nan
    if isinstance(x, (int, float, np.number)): return float(x)
    s = str(x)
    for ch in ("\u00a0", "\u2009", "\u202f"): s = s.replace(ch, " ")
    s = s.strip()
    if s in NIL: return 0.0                    # em / en dash = nil
    if s == "" or s.lower() in NA_LK: return np.nan   # NM, n.a., #N/A N/A
    neg = False
    if s.startswith("(") and s.endswith(")"): neg, s = True, s[1:-1].strip()
```

```
s = s.replace("\u2212", "-").replace("\u2013", "-")
if s.startswith("-"): neg, s = True, s[1:].strip()
pct = s.endswith("%")
s = s.rstrip("%").replace(",", "").replace(" ", "")
try: v = float(s)
except ValueError: return np.nan
v = -v if neg else v
return v / 100 if pct else v
```

Before	After	Note
(10,546.0)	−10,546.0	A naive strip of brackets and commas returns +10,546 and loses the sign
(2\xa0550.0)	−2,550.0	Non-breaking space plus parentheses
−40.0	−40.0	Naive float() raises ValueError on the Unicode minus
-17.7%	−0.177	Percent handled; also "—" returns 0.0 and "NM" returns NaN

Q3. Duplicates

Detected with duplicated() on the label plus all eight values: Net Revenue is identical in label and in every year (rows 17 and 19), and Premium repeats with identical values. It is a true duplicate rather than a legitimate repeat because the product block foots to the total only when Premium is counted once (FY2017: 7,461 + 12,866 + 678 + 875 = 21,880).

Q4. Label normalisation

Whitespace is stripped and collapsed, case is standardised, "&" becomes "and" and British and American spellings are unified. Judgement calls:

- "Digital B2B (eB2B)", "eB2B / Digital" and "Digital B2B" are one channel.
- "C. & E. Europe" equals "Central & Eastern Europe", and "Asia Pacific" equals "Asia-Pacific".
- AMEE equals "Africa, Middle East & Eastern Europe" (confirmed by the REV BY GEO footnote).
- The old "Africa & Middle East" is **not** the same as AMEE. It is a different perimeter and is handled through the mapping table.

Q5. Wide to long

df.melt(id_vars=[...], var_name="fiscal_year", value_name="value") gives one row per line item per fiscal year. A single key (line item and year) then drives every lookup, and Power BI gets one fact table it can slice and filter.

Q6. Corrections

Export & Other FY2019 is divided by 1,000, and excise is tagged with sign −1 (deduction). Both are recorded in the data-quality log.

Q7. Reconciliation checks

Tolerance: EUR 1.5 mn for EUR mn lines and 0.01 for EPS.

Check	Result
Revenue by channel = Net Revenue (after the unit fix)	Ties in all 8 years
Total Revenue − Excise = Net Revenue	Ties in all 8 years

Check	Result
Gross Profit = Net Revenue − Cost of Sales	Ties in all 8 years
EBIT = GP + Other Income − Opex − Impairment	Ties in all 8 years
EBT = EBIT + associates + interest income − interest expense − other finance	Ties in all 8 years
Net Income Attributable = Net Income − Minority Interest	Ties in all 8 years
EPS = Net Income Attributable / weighted average basic shares	Within 0.01 in all 8 years (largest gap 0.005, rounding)

Before the unit fix, the FY2019 channel sum fails by 1,958,040. Discontinued operations NaN is treated as 0 only inside the check, not in the data.

Q8. Export

The notebook writes to the clean folder: master_clean_long.csv (dataset, section, label, fiscal_year, value, placeholder, derived, alloc_dependent, code, group, sign, is_subtotal), is_clean_long.csv, opex_clean_long.csv, geo_restated_long.csv, dim_line_item.csv, dim_region_mapping.csv, data_quality_log.csv and analysis_tables.xlsx. Line item and year drive the lookup, group and sign drive filtering, and source_sheet keeps traceability.

Stage 3: Master P&L with Lookup Functions (Excel)

Q1. Lookup formula

The LOOKUP sheet (cell B6) retrieves any value from the DATA sheet, which holds the cleaned long table plus a key column (M) built as dataset | label | year:

```
=IFERROR(INDEX(DATA!$E:$E,
  MATCH("income_statement|"&B3&"|"&B4, DATA!$M:$M, 0)),
  "Select both inputs")
```

- **MATCH(..., DATA!$M:$M, 0)** finds the row whose key equals the dataset, the selected line item (B3) and the selected year (B4). The 0 forces an exact match.
- **INDEX(DATA!$E:$E, row)** returns the value column (E) at that row.
- **IFERROR(..., "Select both inputs")** shows a friendly message instead of an error.

The MASTER_PL cells use the same pattern with the row label cleaned inside the formula, TRIM(SUBSTITUTE($A5,CHAR(160)," ")), and show "n.a." for blanks.

Q2. Two-way lookup

B3 (line item) and B4 (fiscal year) are Data Validation dropdowns fed by the named ranges Itemlist and Yearlist. Changing the two inputs returns any item for any year. Checks: Net Revenue and 2024 gives EUR 27,640 mn; Basic EPS and 2020 gives −0.34.

Q3. Dictionary lookup and the two missing codes

Statement group and sign convention are pulled from the dictionary with INDEX/MATCH on the cleaned label. Two lines have no code in the dictionary: **Distribution & Logistics** and **Share of Net Profit of Associates & JVs** (LOOKUP TABLES rows 17 and 23). Wrapping the lookup in IFERROR returns "Not in dictionary" instead of #N/A. In the Python dimension table the two lines get provisional codes (IS_DIST, IS_ASSOC) with a code_assigned flag,

and six items are not in the dictionary at all (the four ratio rows and the two share-count rows), which are treated as ratio or per-share items.

Q4. Text functions and exact match

SUBSTITUTE(x, CHAR(160), " ") removes non-breaking spaces. TRIM alone does not, because it only removes ordinary spaces (CHAR(32)); after SUBSTITUTE, TRIM collapses the rest. Labels also need "&" changed to "and" and one spelling. Exact match (0) matters because an approximate match can silently return a neighbouring item with no error, while an exact match fails loudly.

Q5. Restated regions

Mapping with allocation: **Europe = Western Europe + 60% of Central & Eastern Europe; AMEE = Africa & Middle East + 40% of Central & Eastern Europe**. Americas, Asia-Pacific and Head Office map one to one. The REGION sheet applies this to FY2017 to FY2020 (EUR mn):

EUR mn	FY2017	FY2018	FY2019	FY2020	FY2021	FY2022	FY2023	FY2024
Europe	10,296	10,449	10,988	9,347	9,983	11,868	12,582	12,383
Americas	5,645	5,820	6,232	5,068	5,770	7,303	7,786	7,850
AMEE	3,882	4,000	4,281	3,313	3,883	4,564	4,232	3,593
Asia-Pacific	2,319	2,472	2,757	2,248	2,589	3,437	3,949	4,146
Head Office & Elim.	−262	−271	−288	−256	−285	−322	−339	−332
Total = Net Revenue	**21,880**	**22,470**	**23,970**	**19,720**	**21,940**	**26,850**	**28,210**	**27,640**

The total equals Net Revenue in every year (check row = 0). The restated series supports total, Americas, Asia-Pacific and Head Office comparisons across all eight years. It cannot give exact pre-FY2021 levels for **Europe** and **AMEE**: the source never applied the 60/40 split to earlier years, so those two are estimates and are flagged as allocation-dependent.

Q6. Why lookups are safer than pasted values

When next quarter's data is loaded, a lookup-driven report updates and keeps its audit trail, whereas pasted values go stale and hide copy errors. Exact keys also fail loudly instead of quietly pulling a wrong number.

Stage 4: Trend Analysis

Q1. Year-over-year growth

	FY18	FY19	FY20	FY21	FY22	FY23	FY24
Net revenue	2.7%	6.7%	−17.7%	11.3%	22.4%	5.1%	−2.0%
Gross profit	3.3%	7.7%	−22.0%	14.2%	13.3%	6.2%	0.8%
Operating profit (EBIT)	2.4%	8.2%	−79.2%	242.5%	−3.4%	−46.9%	100.4%

	FY18	FY19	FY20	FY21	FY22	FY23	FY24
Net income attributable	4.5%	6.5%	−109.8%	n.m.	−17.2%	−140.7%	n.m.

When the prior year is negative a percentage has no meaning, so the table shows **n.m.** and the move is stated in euros: net income attributable went from a loss of EUR 195 mn (FY2020) to a profit of EUR 1,226 mn (FY2021), and from a loss of EUR 413 mn (FY2023) to a profit of EUR 949 mn (FY2024).

Q2. CAGR FY2017 to FY2024

Net revenue grew at **3.4% a year** (EUR 21,880 mn to EUR 27,640 mn). Net income attributable **fell 8.6% a year** (EUR 1,787 mn to EUR 949 mn). They differ because costs grew faster than sales (operating expenses 4.3% a year against 3.4%), and finance costs, impairments and the Russia exit took the rest. The net revenue CAGR is the more honest headline, because the FY2024 net income end-point is only a partial recovery after the FY2023 loss and is sensitive to one-off items.

Q3. Indexed series (FY2017 = 100)

FY2024 index: net revenue **126.3**, gross profit **120.2**, total operating expenses **134.2** (excluding impairment).

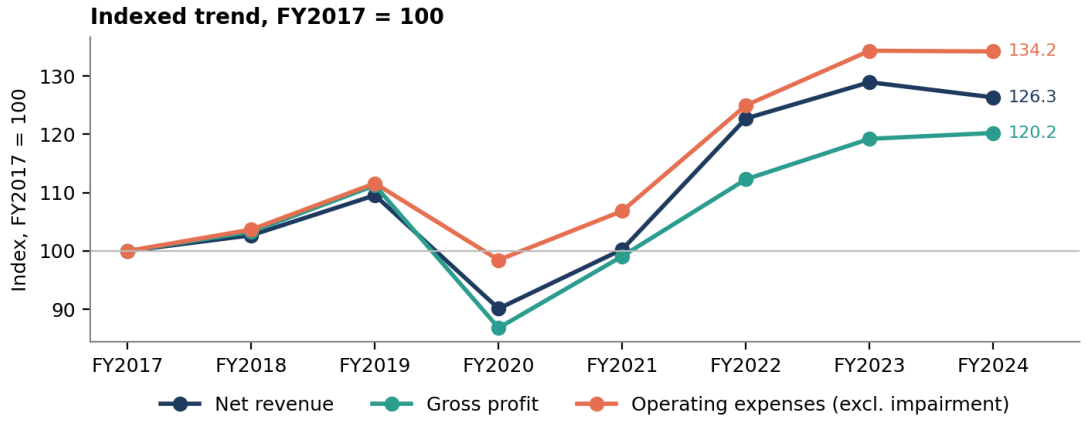


Figure 1: Net revenue, gross profit and operating expenses, FY2017 = 100 (source: RATIOS sheet)

Cost of sales grew faster than sales (gross profit lags revenue), and operating expenses grew faster still, so margin was squeezed from both sides.

Q4. Best and worst years

	Best	Worst
Net revenue	FY2023 (EUR 28,210 mn)	FY2020 (EUR 19,720 mn)
Operating profit	FY2019 (EUR 3,288 mn)	FY2020 (EUR 684 mn)
EPS	FY2019 (EUR 3.46)	FY2023 (EUR −0.72)

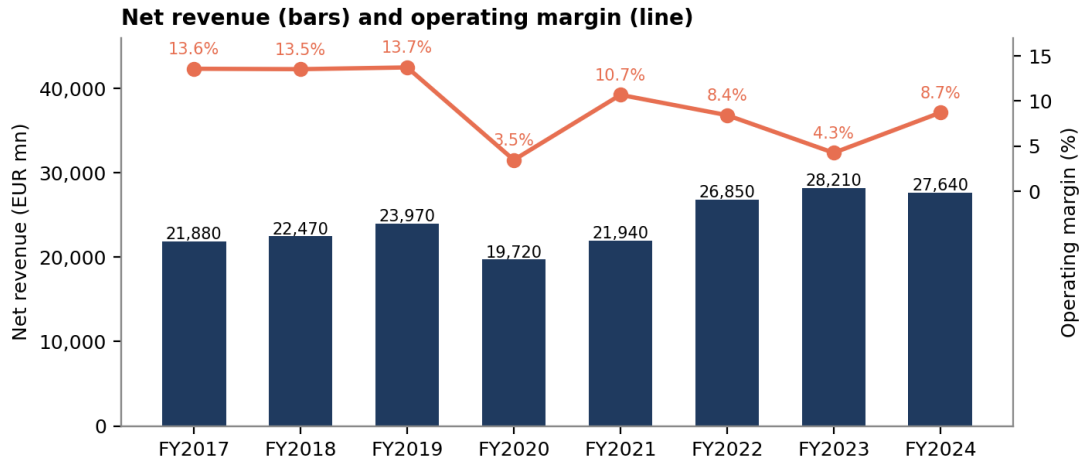


Figure 2: Net revenue (EUR mn) with operating margin overlay (source: RATIOS sheet)

FY2020: bars and restaurants closed (On-Trade fell from EUR 9,740 mn to EUR 5,920 mn) and a EUR 900 mn impairment was booked. **FY2023:** a EUR 1,050 mn impairment, higher finance costs and the Russia exit (EUR −310 mn from discontinued operations) produced a net loss even though net revenue was at its peak.

Q5. Revenue versus operating expenses; operating leverage

Operating expenses grew about **4.3% a year** against **3.4%** for net revenue. Operating leverage was therefore negative: each extra euro of sales brought less profit, and a fall in sales hurts profit more than proportionally. Over FY2019 to FY2024 the gap is the same: operating expenses +20.3% against net revenue +15.3%.

Q6. Base effect

FY2021 operating profit growth of **+242.5%** comes off the depressed FY2020 base of EUR 684 mn. Presented fairly, it should be shown with levels (EUR 684 mn to EUR 2,343 mn) and against the FY2019 peak of EUR 3,288 mn, which shows it is still 29% below. FY2024 (+100.4%) has the same issue against FY2023's EUR 1,201 mn, which was itself depressed by the impairment.

Q7. Net revenue per hectolitre and the volume / price-mix split

EUR per hl	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24
Net revenue per hl	100.4	96.2	99.3	88.7	94.5	111.6	122.2	122.1

FY2017 was n.a. in the source and is derived as 21,880 / 218.0 = 100.4. Volume was 241.4 mn hl in FY2019 and 226.4 mn hl in FY2024.

- Volume effect = (226.4 − 241.4) × 99.30 = **EUR −1,489 mn**
- Price / mix effect = (122.08 − 99.30) × 226.4 = **EUR +5,159 mn**
- Total = EUR 3,670 mn = 27,640 − 23,970 (check = 0)

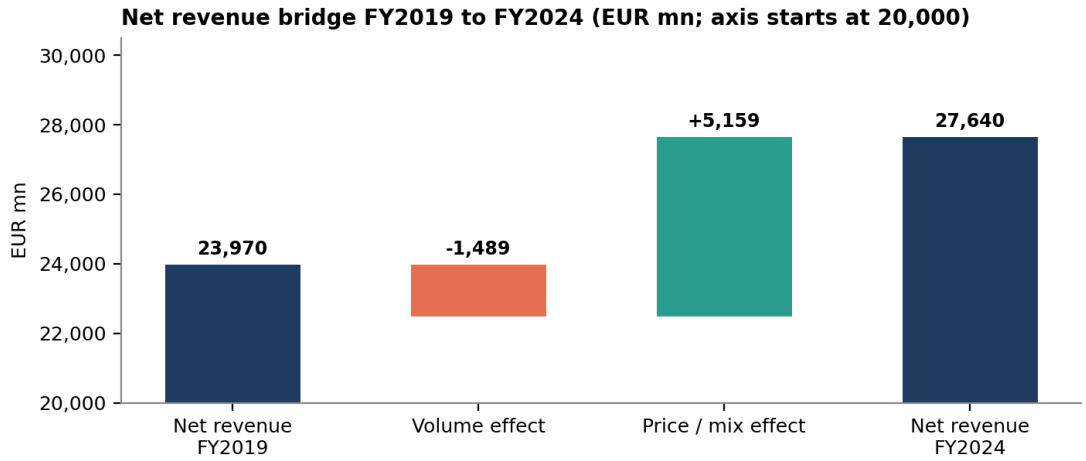


Figure 3: Net revenue bridge FY2019 to FY2024, EUR mn (source: RATIOS rows 42 to 54)

All of the growth came from price and mix; volume was lower than in FY2019.

Stage 5: Profitability and Margin Analysis

Q1. Margins on net revenue

	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24
Gross margin	51.8%	52.1%	52.6%	49.9%	51.2%	47.4%	47.9%	49.3%
Operating margin	13.6%	13.5%	13.7%	3.5%	10.7%	8.4%	4.3%	8.7%
Net margin	8.2%	8.3%	8.3%	−1.0%	5.6%	3.8%	−1.5%	3.4%

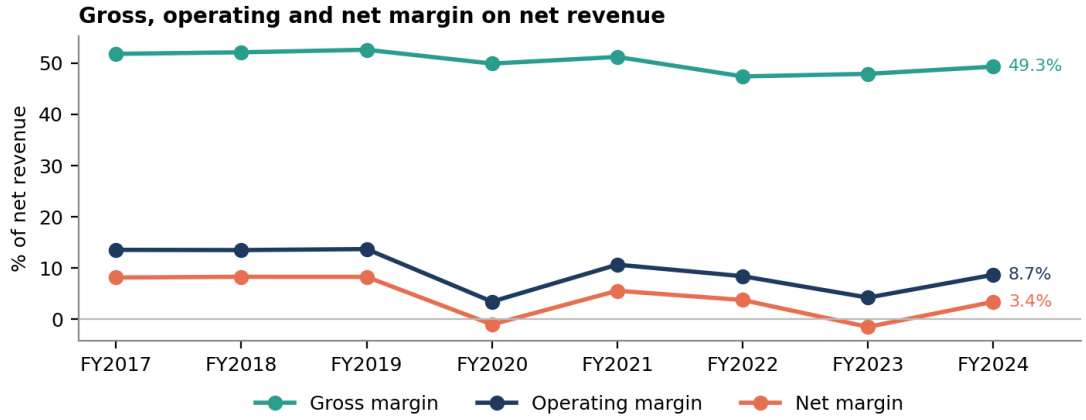


Figure 4: Gross, operating and net margin on net revenue (source: RATIOS rows 10 to 12)

Q2. Gross margin on total revenue

Including excise, gross margin is **43.9%** (FY2017) and **42.0%** (FY2024), which is **7.9** and **7.3 points** lower than on net revenue. I would show management the net revenue basis: excise is a pass-through tax, and the larger denominator understates real performance and mixes in tax changes that management cannot control.

Q3. Operating margin change FY2019 to FY2024 (13.7% to 8.7%, −5.0 points)

Line	Change in % of net revenue
Gross margin	−3.3 pp
Other income	0.0 pp

Line	Change in % of net revenue
Personnel	−0.3 pp
Marketing and selling	+0.5 pp
Distribution and logistics	−1.4 pp
General and administrative	−0.3 pp
Depreciation and amortisation	−0.2 pp
Impairment	0.0 pp
Total	**−5.0 pp**

Gross margin did the most damage (−3.3 points): cost of sales rose from 47.4% to 50.7% of net revenue. Among operating expenses, distribution and logistics was worst (−1.4 points).

Q4. Operating expenses as % of net revenue

% of net revenue	FY2017	FY2024	Change
Personnel	11.7%	11.8%	+0.1 pp
Marketing and selling	11.8%	11.4%	−0.4 pp
Distribution and logistics	6.2%	7.7%	+1.6 pp
General and administrative	3.1%	3.3%	+0.2 pp
Depreciation and amortisation	5.9%	6.8%	+0.9 pp
Total (excl. impairment)	**38.6%**	**41.0%**	**+2.4 pp**

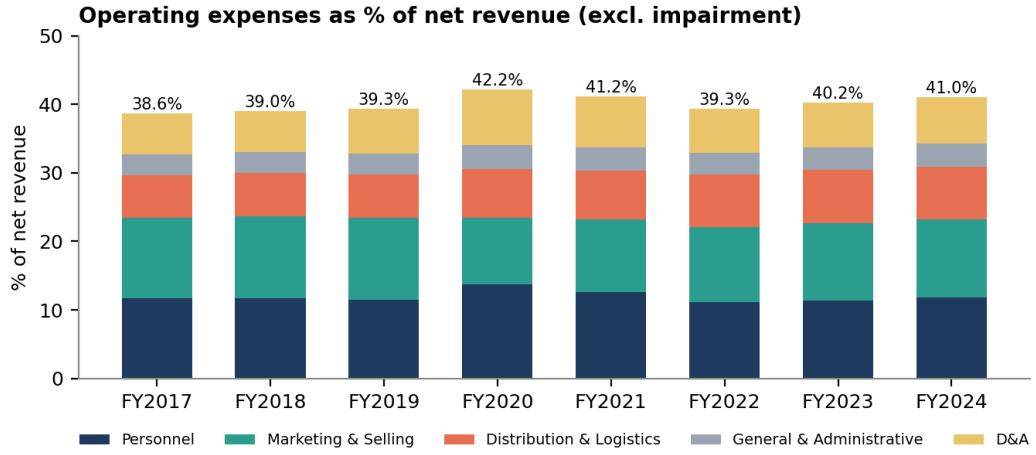


Figure 5: Operating expenses as % of net revenue (source: RATIOS rows 30 to 35)

Most volatile in EUR: marketing and selling (EUR 2,860 mn in FY2019 to EUR 1,930 mn in FY2020, −32.5%). **Most fixed-like:** personnel (−1.8% in FY2020) and depreciation and amortisation (+1.9%). A fixed cost looks volatile as a percentage of sales because the denominator moves: personnel rose from 11.5% to 13.7% of net revenue in FY2020 while its euro amount barely changed. In a downturn, margin therefore falls faster than sales.

Q5. Why net margin falls further than operating margin (FY2021 to FY2023)

Operating margin fell 6.4 points (10.7% to 4.3%) and net margin fell 7.1 points (5.6% to −1.5%). Below EBIT:

- Net finance lines (interest, other finance and associates) went from a cost of EUR 408 mn to EUR 1,008 mn: about **−1.7 points**.

- Discontinued operations (the Russia exit) were EUR −310 mn in FY2023: about **−1.1 points**.
- Lower tax (EUR 524 mn to EUR 91 mn) cushioned it by about **+2.1 points**, and minority interest by about +0.1.

Q6. EPS bridge, FY2022 to FY2024

EPS = net income attributable / weighted average basic shares: FY2022 = 1,015 / 575.2 = **1.76**; FY2024 = 949 / 561.4 = **1.69**, a change of −0.07. The profit effect is (949 − 1,015) / 575.2 = **−0.11**; the share-count effect (fewer shares) is **+0.04**. Lower profit reduced EPS, and buybacks offset about a third of it.

Q7. Effective tax rate above 100%

FY2020: tax of EUR 272 mn on income before taxes of EUR 197 mn = **138.1%**. It arises when a large charge (here the EUR 900 mn impairment) cuts pre-tax profit but is not tax-deductible, while tax is still charged on the rest. Excluding the impairment the rate is 272 / 1,097 = 24.8%, in line with the normal 26%. I would show EBIT before impairment (EUR 1,584 mn, 8.0% margin) next to the reported 3.5%, and label the headline tax rate n.m.

Q8. Best single ratio

Underlying operating margin (EBIT before impairment / net revenue): 13.6% (FY2017), 13.7% (FY2019), 8.0% (FY2020), 8.0% (FY2023), 8.7% (FY2024). It excludes pass-through excise, financing, tax and discontinued operations, and strips one-off impairments. It shows the recovery has stopped about five points below the FY2019 level. Caveat: IFRS 16 (from FY2019) raised depreciation.

Stage 6: Segment and Geographic Analysis

Q1. Revenue mix

Product category	FY2017	FY2024	Change
Premium	34.1%	41.2%	+7.1 pp
Mainstream	58.8%	47.5%	−11.3 pp
Low & No-Alc.	3.1%	6.1%	+3.0 pp
Beyond Beer	4.0%	5.2%	+1.2 pp

Channel	FY2017	FY2024	Change
On-Trade	42.0%	37.7%	−4.3 pp
Off-Trade	49.7%	51.6%	+1.9 pp
Digital B2B	0.5%	6.6%	+6.0 pp
Export & Other	7.8%	4.2%	−3.6 pp

Brand	FY2017	FY2022	Change
Heineken	21.2%	24.6%	+3.4 pp
Amstel	6.1%	5.7%	−0.4 pp
Desperados	3.4%	4.2%	+0.8 pp
Tiger	3.9%	4.9%	+1.0 pp

Brand	FY2017	FY2022	Change
Other brands	65.4%	60.6%	−4.8 pp

The mix moved towards premium: Premium gained 7.1 points while Mainstream lost 11.3. Off-trade and digital gained at the expense of on-trade and export. Among brands, Heineken gained 3.4 points of share.

Q2. On-trade versus off-trade

On-trade share fell from 42.0% to 37.7% (EUR 9,180 mn to EUR 10,410 mn, +13.4%), while off-trade rose from 49.7% to 51.6% (EUR 10,870 mn to EUR 14,250 mn, +31.1%). The business has shifted towards retail and away from bars and restaurants. This was driven by FY2020, when on-trade share fell from 40.6% (FY2019) to 30.0% and off-trade jumped to 61.9%. By FY2024 on-trade had regained 7.7 of the 10.6 points lost (about 73%), so most but not all of the pandemic shift reversed. In euros on-trade (EUR 10,410 mn) is above FY2019 (EUR 9,740 mn), but its share (37.7%) is still below FY2019 (40.6%).

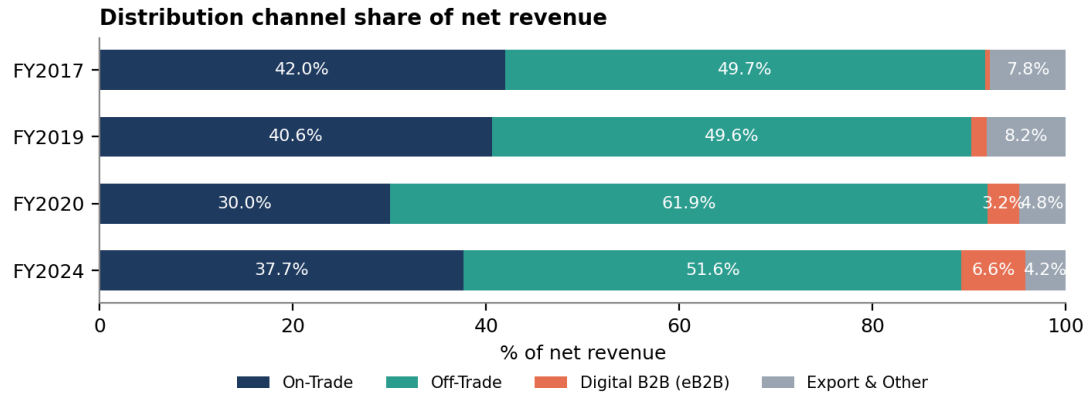


Figure 6: Distribution channel share of net revenue (source: SEGMENTS sheet)

Q3. Digital B2B growth

It grew **1,416.7%**, but from a tiny base of EUR 120 mn. Report alongside it the absolute increase of **EUR 1,700 mn** (EUR 120 mn to EUR 1,820 mn), which is 29.5% of the total net revenue increase of EUR 5,760 mn, and its share of net revenue (0.5% to 6.6%).

Q4. Brand revenue after FY2022

Footnote (1) of the extract says that from FY2023 brand performance is disclosed by volume only, so brand revenue is no longer reported (#N/A N/A in FY2023 and FY2024). I kept those cells as missing (NaN), not zero, and compared brands for FY2017 against FY2022 only. The table and chart state the end year, so the reader knows the comparison stops at FY2022.

Q5. Regional contribution to the change in net revenue, FY2019 to FY2024 (restated basis)

Net revenue rose EUR 3,670 mn (EUR 23,970 mn to EUR 27,640 mn).

Region	Contribution (EUR mn)	Share of total change	Growth
Americas	+1,618	44.1%	26.0%
Europe	+1,395	38.0%	12.7%
Asia-Pacific	+1,389	37.8%	50.4%
AMEE	−688	−18.7%	−16.1%

Region	Contribution (EUR mn)	Share of total change	Growth
Head Office & Eliminations	−44	−1.2%	n.m.
Total	**+3,670**	**100.0%**	**15.3%**

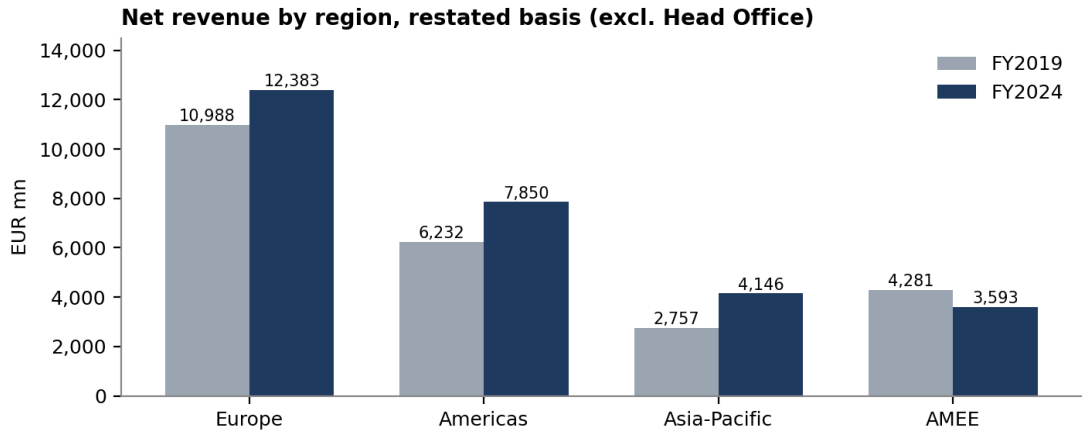


Figure 7: Net revenue by region, FY2019 vs FY2024, restated basis (source: REGION sheet)

The Americas contributed most of the growth (EUR 1,618 mn) and AMEE was the only material decline (EUR −688 mn). A contribution analysis is more useful than growth rates because it shows how much each region matters in euros and the contributions add up to the total change. Asia-Pacific has the highest growth rate (50.4%) but starts from a smaller base, so it adds less than the Americas. Europe grew only 12.7% yet added EUR 1,395 mn because of its size. Europe and AMEE FY2019 values use the approximate 60/40 split, so those two contributions are estimates.

Q6. Net revenue per employee and per hectolitre

	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24
Net revenue per employee (EUR ’000)	273.1	262.5	279.0	233.7	266.5	315.0	314.4	317.7
Net revenue per hl (EUR)	100.4	96.2	99.3	88.7	94.5	111.6	122.2	122.1

Both fell in FY2020 (pandemic) and rose strongly from FY2022. Over the period net revenue per employee grew **16.3%** and net revenue per hl grew **21.6%**. Read together, the productivity gain is a price story, not a volume story: headcount grew 8.6% (80,120 to 87,010) and volume only 3.9% (218.0 to 226.4 mn hl), so beer sold per employee fell about 4.4%. **Limits:** revenue per employee mixes output with pricing, uses year-end headcount rather than an average, and is affected by outsourcing and acquisitions. Revenue per hl measures price and mix, not efficiency, and mix shifts can move it without any change in pricing power. Neither measures profit.

Stage 7: Professional Formatting and Excel Charts

Q1. Formatting of the master P&L

Units are named in the sheet header (“EUR mn, except per-share data”). Amounts use one number format, #,##0;(#,##0);"-", so negatives appear in brackets and nil shows as a dash; EPS and DPS use 0.00;(0.00) and share counts use #,##0.0, so decimals line up by row type. Every value is a real number returned by a lookup, so no text-formatted numbers remain. The header

row is bold, column A is wide enough for the longest label, and the panes are frozen so line items stay in view while scrolling.

Q2. Conventions for inputs, formulas and links

Blue font marks hard-coded inputs (the LOOKUP dropdowns, which are blue on yellow, the 60/40 allocation, the reported regional figures and the volume row), black marks formulas on the same sheet, and green marks values pulled from another sheet. A person opening the model six months later can see at once which cells they may change, which are calculated and where a number comes from, and it prevents formulas being overwritten by accident.

Q3. Conditional formatting

A three-colour scale (red at the low end) is applied to the gross margin and operating margin rows on the RATIOS sheet, each scaled to its own range. In under five seconds a reader sees that gross margin was weakest in FY2022 (47.4%), that operating margin was weakest in FY2020 (3.5%) and again in FY2023 (4.3%), that FY2017 to FY2019 were the strong years, and that FY2024 has recovered only part of the way (operating margin 8.7% against 13.7% in FY2019).

Q4. Charts and why each type

Chart	Type	Why
Net revenue with operating margin (Figure 2)	Combo: columns and line	Two measures in different units (EUR mn and %); shows revenue recovered but margin did not
Gross, operating and net margin (Figure 4)	Line	Three percentage series over time on one axis; shows direction and dips
Operating expenses as % of net revenue (Figure 5)	Stacked column	Shows composition and the total (38.6% in FY2017 to 41.0% in FY2024)
Net revenue by region, FY2019 vs FY2024 (Figure 7)	Clustered bar	Side-by-side comparison of categories; AMEE is the only region that fell
Net revenue bridge (Figure 3)	Waterfall	Bridges two totals: EUR 23,970 mn, volume −1,489, price/mix +5,159, EUR 27,640 mn

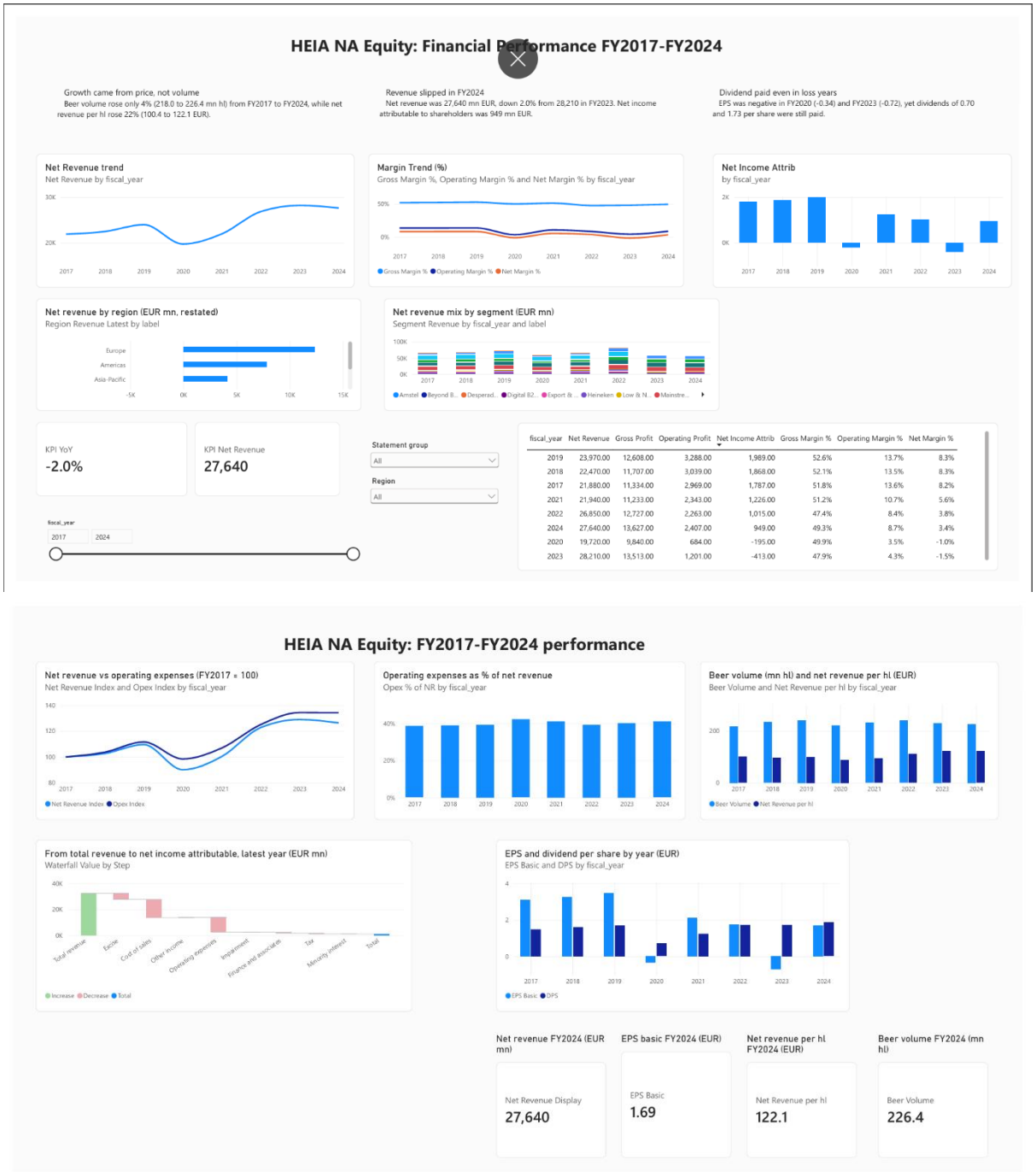
Q5. What was left off the charts

Gridlines were removed, data labels are kept only where they help (whole numbers for revenue, one decimal for percentages), and the dual axis appears only in the combo chart where the units genuinely differ. No 3-D effects or decorative colours were used. The regional chart excludes Head Office and Eliminations because it is a negative reconciling item, not a market. The waterfall uses an invisible base series so only the bridging steps show, and its legend is removed.

Q6. Summary sheet

The SUMMARY sheet holds only what management should see first: a table of eight metrics (net revenue, EBIT, net income attributable, gross, operating and net margin, operating expenses as % of net revenue, net revenue per hl) for FY2017, FY2019 (the pre-pandemic benchmark) and FY2024 with the change from FY2019, plus two charts (net revenue with operating margin, and the bridge). Net revenue is up EUR 3,670 mn since FY2019, but operating margin is down 5.0 points, operating expenses are up 1.7 points as a share of net revenue, and net revenue per hl is up EUR 22.8.

Stage 8: Power BI Dashboard



Q1. Data model

I loaded **one tidy fact table**, master_clean_long (608 rows, 12 columns: dataset, section, label, fiscal_year, value, placeholder, derived, alloc_dependent, code, group, sign, is_subtotal), one row per line item per fiscal year. It holds four datasets (income statement 304 rows, segment 128, opex detail 136, restated regions 40), so one table reaches every number. The statement group, sign and code from the dictionary, and the restated regions, were already joined in Python, so no relationships were needed in this build. The line-item and region-mapping dimension tables (dim_line_item.csv, dim_region_mapping.csv) are exported with the clean data and can be added as dimensions if more slicers are needed. One table with one grain avoids join ambiguity and keeps every visual consistent.

Q2. DAX measures versus Python

Seven measures are written in DAX:

- Net Revenue, Gross Profit, Operating Profit and Net Income Attrib, each CALCULATE(SUM(master_clean_long[value]), dataset = "income_statement", label = "...").
- Gross Margin %, Operating Margin % and Net Margin %, each DIVIDE(profit measure, [Net Revenue]).

Python does the row-level work that never changes: text-to-number conversion, unit and sign fixes, duplicates, the regional allocation and the reconciliation checks. DAX holds what must react to slicers. Margins are written as a ratio of sums, not an average of yearly ratios, so they stay correct for any year selection. Cleaning is done once and tested; measures recalculate for whatever the user selects.

Q3. Layout

Page 1 reads top to bottom. A title text box sits at the top. Below it on the left is the net revenue trend line, with the fiscal-year slicer under it; in the centre is the margin-trio line chart (gross, operating and net margin); on the right is a column chart of net income attributable. A table at the bottom lists net revenue, gross profit, operating profit, net income attributable and the three margins by year. A reader with sixty seconds gets scale (revenue), quality (margins), outcome (earnings) and then the detail, in that order.

Q4. Most useful visual: CFO versus external analyst

They are not the same. For the **CFO** the most useful visual is the margin-trio line chart read with the table beneath it: the gap between gross and operating margin shows where cost leaks, and the table gives the euro figures to act on. For an **external equity analyst** it is the net income attributable chart together with net margin, because it shows earnings quality after financing, tax and discontinued items, which drive valuation. The CFO needs a diagnosis; the analyst needs a trend.

Stage 9: Insights and Storytelling

Q1. Peak net revenue and the eight-year change

Net revenue peaked in **FY2023 at EUR 28,210 mn**. From FY2017 to FY2024 it rose **EUR 5,760 mn** (EUR 21,880 mn to EUR 27,640 mn), a **CAGR of 3.4%**.

Q2. Gross margin range

Highest in **FY2019 (52.6%)**, lowest in **FY2022 (47.4%)**: a spread of **5.2 percentage points**.

Q3. Largest falls in operating profit

The largest was **FY2020: EUR −2,604 mn** (EUR 3,288 mn to EUR 684 mn). The second largest was **FY2023: EUR −1,062 mn** (EUR 2,263 mn to EUR 1,201 mn). FY2020 was a revenue shock (net revenue −17.7%) plus a EUR 900 mn impairment. FY2023 was different: revenue still grew 5.1%, and EBIT before impairment was EUR 2,251 mn against EUR 2,263 mn, so EUR 1,050 mn of the EUR 1,062 mn fall is the impairment.

Q4. Operating expense line that rose most as a share of net revenue

Distribution and logistics, from 6.2% (FY2017) to 7.7% (FY2024): **+1.6 points**. Next is depreciation and amortisation (5.9% to 6.8%, +0.9 points).

Q5. Fastest growth

Channel: **Digital B2B, +1,416.7%** (EUR 120 mn to EUR 1,820 mn), an absolute increase of **EUR 1,700 mn**. Region: **Asia-Pacific, +78.8%** FY2017 to FY2024 (EUR 2,319 mn to EUR 4,146 mn, +EUR 1,827 mn). Americas grew 39.1%, Europe 20.3% and AMEE fell 7.4%. Europe and AMEE depend on the approximate allocation for FY2017.

Q6. EPS

Highest in **FY2019 (EUR 3.46)**, lowest in **FY2023 (EUR −0.72)**: a range of **EUR 4.18**. A dividend was still paid in the lowest year: DPS of **EUR 1.73** (the same as FY2022), about EUR 986 mn in total against a net loss attributable of EUR 413 mn. A dividend was also paid in the FY2020 loss year (EUR 0.70). It shows management treats the dividend as sticky and looks through one-off charges, but in FY2023 it was not covered by earnings and cannot be repeated for long.

Q7. Was FY2022 to FY2023 a revenue, margin or below-the-line problem?

Mainly a **margin problem**, made worse **below the line** in FY2023. It was not a revenue problem: net revenue grew 28.6% from FY2021 to FY2023 (EUR 21,940 mn to EUR 28,210 mn). In FY2022 cost of sales rose 31.9% against revenue +22.4%, so gross margin fell from 51.2% to 47.4% and operating margin from 10.7% to 8.4%, then 4.3% in FY2023. In FY2023 net margin fell to −1.5% because of net finance costs of EUR 1,008 mn (EUR 408 mn in FY2021), discontinued operations of EUR −310 mn and the EUR 1,050 mn impairment.

Q8. Three insight cards

> **Sales are back, profit is not**
> Net revenue is EUR 3.7 bn higher than in FY2019, but operating profit is EUR 0.9 bn lower.

> **Growth came from price, not more beer**
> Higher prices and a richer mix added EUR 5.2 bn, while lower volume cost EUR 1.5 bn.

> **Costs ate the gain**
> For every EUR 100 of sales the brewer now keeps EUR 8.70 as operating profit, against EUR 13.70 in FY2019.

Memo to the CFO

To: Chief Financial Officer **From:** [Meghna Tripathi], Finance Analyst **Date:** 31 October 2026

Subject: HEIA NA Equity, FY2017 to FY2024: what the record says and what to watch

What the eight-year record says. Net revenue grew 3.4% a year to EUR 27,640 mn (FY2024). Since FY2019 all of the growth has come from price and mix (+EUR 5,159 mn) while volume fell (−EUR 1,489 mn; 226.4 mn hl against 241.4 mn hl). Profit did not follow: operating margin is 8.7% against 13.7% in FY2019, and net income attributable is EUR 949 mn against EUR 1,989 mn. Of the 5.0 points of operating margin lost, 3.3 came from gross margin (cost of sales rose from 47.4% to 50.7% of net revenue) and 1.4 from distribution and logistics. FY2020 was a demand shock (net revenue −17.7%, EUR 900 mn impairment). FY2023 was mostly a EUR 1,050 mn impairment on top of higher finance costs (EUR 1,008 mn) and the Russia exit (EUR −310 mn). The mix has moved to premium and off-trade, and Asia-Pacific and the Americas drove the growth while AMEE declined (EUR −688 mn since FY2019).

Two numbers to watch next year.

1. **Operating margin before impairment (8.7% in FY2024).** A move back above 10% would show cost pressure is easing; a fall would show price increases are no longer covering costs.
2. **Beer volume (226.4 mn hl against 241.4 mn hl in FY2019).** Price-led growth depends on pricing power continuing to hold. Falling volume with no further price gain would put revenue itself at risk.

Recommendation. Focus cost action on cost of sales and on distribution and logistics, which together explain 4.7 of the 5.0 points of lost operating margin. Recovering one point of gross margin is worth about EUR 276 mn of operating profit on FY2024 net revenue (1% of EUR 27,640 mn). I would also treat the FY2023 dividend (about EUR 986 mn against a loss of EUR 413 mn) as a risk: it was not covered by earnings.

Data note: the extract is synthetic (generated for teaching) and must not be presented as actual company results. All figures trace to the Python clean data, the Excel model and the Power BI file.

Final Reflection

The defect that would have changed my conclusions most is the excise sign. Had excise been added instead of subtracted, FY2017 net revenue would have been EUR 29,700 mn instead of EUR 21,880 mn and gross margin 38.2% instead of 51.8%, so every margin in every year would have been wrong by roughly 13 to 14 points, and the whole profitability story with it. Nothing on the page would have looked broken.

The **impairment** is the runner-up. Left inside operating expenses, FY2020 opex would read 46.8% of net revenue instead of 42.2%, and the FY2023 fall in operating profit would be blamed on operating costs when EUR 1,050 mn of the EUR 1,062 mn fall is the impairment. The **unit error** would have corrupted the FY2019 channel mix but not the totals, and the reconciliation check caught it immediately.

How the three tools worked together. Python (cleaning and reconciliation) produced a dataset that ties to the statement line by line, and the checks made each defect visible instead of hidden. Excel (structure, lookups and formatting) made every number traceable to a cell and let a reader change two inputs to retrieve any figure. Power BI (storytelling) turned the same data into a view a manager can read in a minute. Because the three share one cleaned dataset, the conclusions rest on checked numbers rather than an impression.